LociLand

VCG Project — Project Approval Proposal

Memory Palace
Builder App
Ages 6 – 14

Student: Ángel Pérez Castro	**Platform:** iOS & Android	**Timeline:** 8 weeks	**Stack:** React Native + Firebase

1. Project Overview

LociLand is a cross-platform mobile application (iOS & Android) designed to teach children aged 6 to 14 the Method of Loci — a centuries-old memorisation technique used by memory champions worldwide. Users build interactive "memory palaces": virtual spaces populated with vivid, memorable anchor points tied to the information they wish to recall. The application combines structured pedagogy with cartoon-style visuals, guided practice, and a gamification layer to sustain long-term engagement.

2. Learning Objective & Pedagogy

The Method of Loci works by binding abstract information to concrete spatial locations within a familiar or imagined environment. Research in cognitive science confirms that spatial-associative encoding significantly improves recall performance. LociLand operationalises this process in three stages:

- Encode — the child selects a palace template (house, castle, forest, space station) and places station markers at chosen spots, each linked to a piece of information.
- Visualise — a cartoon guide character walks the child through the palace during creation, reinforcing the spatial-visual association.
- Retrieve — a guided review mode tours the palace station by station, prompting active recall before revealing the answer, with immediate animated feedback.

The UI complexity adapts automatically to age group: the 6–9 interface prioritises large emoji anchors and voice-friendly labels, while the 10–14 interface exposes richer text, statistics, and streak data.

3. Core Features

Feature	Description	Category
Palace Builder	Create named memory palaces from illustrated templates (house, castle, forest, space)	Core
Station System	Add anchor points inside each palace; attach emoji, image, and short text to each	Core
Guided Review	Step-by-step recall quiz that walks children through their stations with instant feedback	Core
Gamification	XP points, daily streaks, achievement badges, and "Memory Master" level progression	Engagement
Auth & Profiles	Firebase email/Google login; age-group selector (6–9 / 10–14) adapting the UI complexity	Auth
Cloud Sync	All palaces and progress stored in Firestore, synced across devices in real time	Backend
Onboarding	Interactive tutorial with animated cartoon guide character teaching the Method of Loci	UX

4. Technical Stack

Technology	Layer	Purpose

React Native + Expo	Frontend / Mobile	Cross-platform iOS & Android from a single codebase
TypeScript	Language	Type safety, fewer runtime errors, better IDE support
Firebase (Auth + Firestore)	Backend / Cloud	Authentication, real-time database, cloud sync
Zustand	State Management	Lightweight global state, simpler than Redux
React Navigation v6	Navigation	Stack and tab navigation between screens
Reanimated 2 + Lottie	Animations	Fluid transitions and cartoon character animations
NativeWind	Styling	Tailwind CSS utility classes adapted for React Native

5. System Architecture

The application follows a client-driven architecture. The React Native frontend manages all UI state locally via Zustand. Firebase handles identity (Auth) and persistence (Firestore), enabling seamless multi-device sync. No custom backend server is required.

Data model summary:

- users/{uid} → profile, age group, XP, level, streak
- users/{uid}/palaces/{palaceId} → name, template, createdAt
- users/{uid}/palaces/{palaceId}/stations/{stationId} → emoji, text, imageUrl, position
- users/{uid}/reviews/{reviewId} → palaceId, score, timestamp

6. Development Timeline (8 Weeks)

Week	Phase	Deliverables
Wk 1	**Foundation & Visual Design**	Theme system, navigation setup, onboarding screens
Wk 2	**Authentication & Profiles**	Firebase Auth, login/register UI, age-group profile
Wk 3	**Palace Creation**	Home screen, palace templates with illustrated assets
Wk 4	**Station System**	Add/edit stations, emoji picker, Firestore sync
Wk 5	**Guided Review Mode**	Recall quiz engine, animated feedback, session history
Wk 6	**Gamification**	XP system, streaks, badges, level progression screen
Wk 7	**Polish & Animations**	Lottie character, transitions, responsive layout fixes
Wk 8	**Testing & Delivery**	Bug fixes, EAS build, demo prep, documentation

7. Design & UX Principles

- Cartoon-first aesthetic: rounded shapes, saturated colour palette (warm yellows, sky blues, coral orange), expressive emoji-based iconography.
- Child-safe typography: Fredoka One (headings) + Nunito (body) — both optimised for readability at small sizes on mobile screens.
- Zero cognitive overload: maximum two interactive actions per screen; progress is always clearly visible.
- Delight-driven feedback: Lottie confetti and character celebrations on every successful recall action.

8. Questions for the Professor

Q1. Are there any changes or adjustments required before implementation begins?